

Ergonomics and Universal Design

Lecture 8

Outline

- What is ergonomics?
- Posture and Movement
- Information and Operation
- Universal Design
- Classroom Activity – Fusion 360:
 - 2D sketching
 - Using constraints
 - Extruding solid bodies
 - Modifying solid bodies
 - Add fillets to solid bodies

What is Ergonomics?

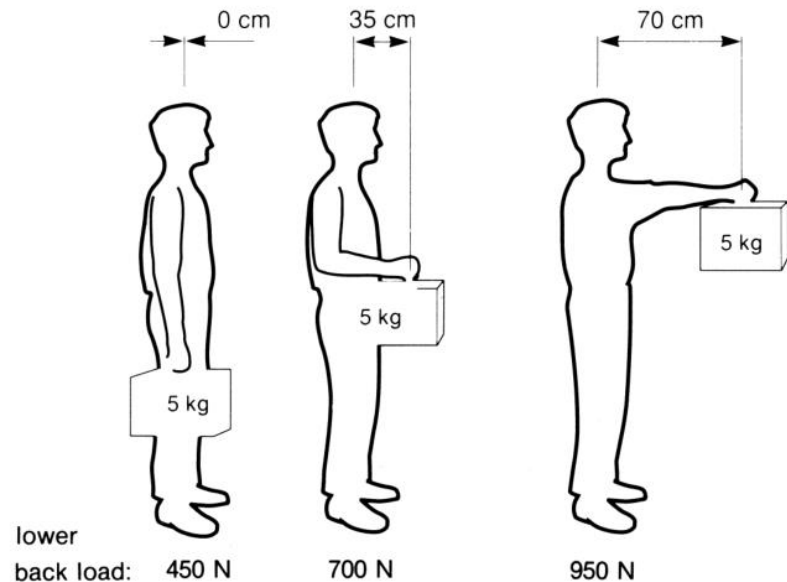
- Field of **ergonomics** (aka human factors) aims to design appliances, technical systems and tasks in such a way as to improve **human safety, health, comfort** and **performance**.
- Developed as a scientific discipline after WW2.
- Ergonomics studies different factors:
 - Body posture and movement
 - Environmental factors
 - Information and operation
 - Work organization

Posture and Movement

- Postures:
 - Sitting,
 - Standing
 - Alternating
- Movements:
 - Lifting
 - Carrying
 - Pulling
 - Pushing
- **Biomechanics** is a discipline that studies mechanical stress on muscles and joints while adopting a posture or making a movement.
- **Anthropometry** is concerned with the size and proportions of human body.
- **Exercise physiology** estimates energy demands on the heart and lungs resulting from muscular effort during movement

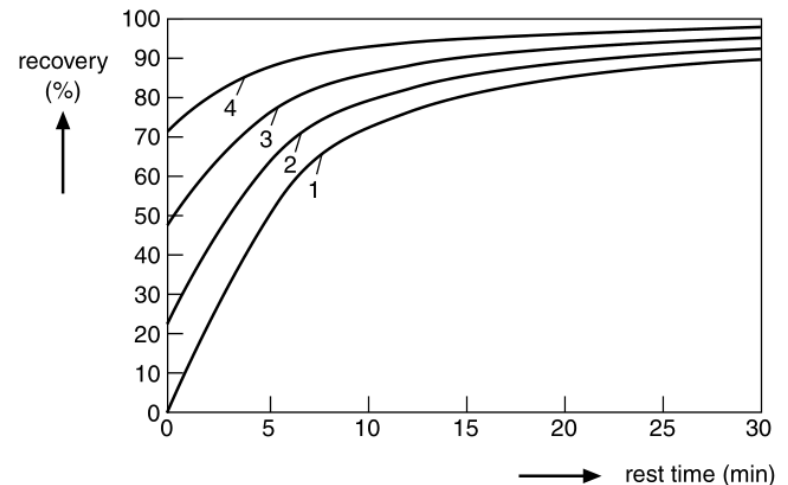
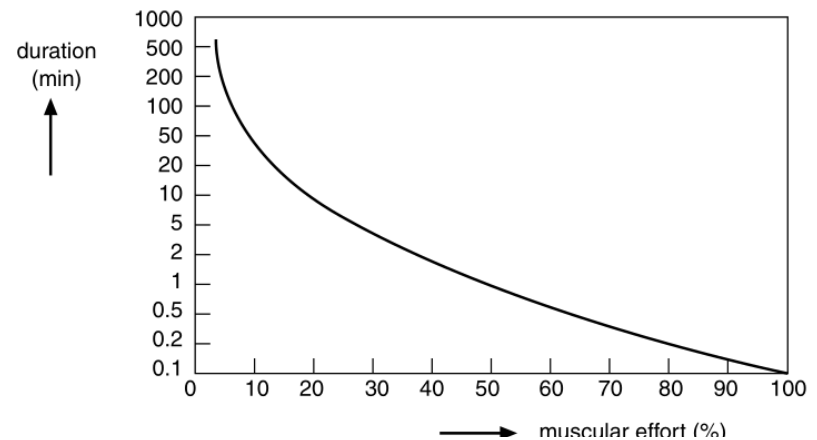
Biomechanical Principles

- Joints must be in a neutral position
 - Raised arms, bent wrists, bent neck, twisted trunk are examples of poor posture.
- Keep the work close to body
 - If the work is too far from the body, arms outstretched and trunk bent over forwards.
- Avoid bending forward
 - Upper half of adult weighs around 40 kgs.
 - The further the trunk is bent, the harder it is for the muscles of the back to maintain the upper body balance.



Biomechanical Principles

- Sudden movements and forces produce peak stresses
 - Thorough preparation is necessary before large forces are exerted
- Limit the duration of any continuous muscular effort
 - The greater the muscular effort, the shorter the time it can be maintained
- Prevent muscular exhaustion
 - Fully exhausted muscle needs to rest for 30 minutes to achieve a 90% recovery.
 - Half exhausted muscles needs 15 minutes to achieve 90% recovery.
- Alternate posture and movements
- Frequent short-breaks are better than a single long one



Physiological Principles

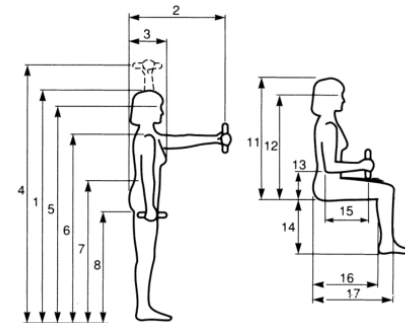
- Limit energy expenditure in a task
 - Task does not exceed 250W (or 0.0145 kcal/min)
 - 80W body at rest
 - Activities less than 250W include writing, ironing, assembling light materials or gentle walk.

<i>Activity</i>	<i>Energy expenditure</i>
Walking while carrying a load (30 kg, 4 km hr ⁻¹)	370 W
Frequent lifting (1 kg, 1 per sec)	600 W
Running (10 km hr ⁻¹)	670 W
Cycling (20 km hr ⁻¹)	670 W
Climbing stairs (30 deg, 1 km hr ⁻¹)	960 W

Anthropometric Principles

- Anthropometric tables
 - Table shows dimensions for unclothed, unshod (add 3-5 cms to account for shoes) persons.
 - Covers around 90% of population
- Take account of differences in body size
 - Control panels consider shortest user
 - Doors consider tallest user

	<i>short</i>	<i>average</i>	<i>tall</i>
<i>Standing</i>			
1. Stature	150.5	167.5	185.5
2. Forward grip reach	65.0	74.3	83.5
3. Chest depth	21.0	25.0	28.5
4. Vertical grip reach	179.0	198.3	219.0
5. Eye height	140.5	156.8	174.5
6. Shoulder height	121.5	136.8	153.5
7. Elbow height	93.0	104.8	118.0
8. Knuckle height	66.0	73.8	82.5
<i>Sitting</i>			
11. Sitting height	79.5	88.0	96.5
12. Sitting eye height	68.5	76.5	84.5
13. Sitting elbow height	18.5	24.0	29.5
14. Popliteal height	35.5	42.0	49.0
15. Elbow-grip length	30.4	34.3	38.7
16. Buttock-popliteal length	43.5	48.8	55.0
17. Buttock-knee length	52.0	58.3	64.5
00. Body weight	44.1	68.5	93.7



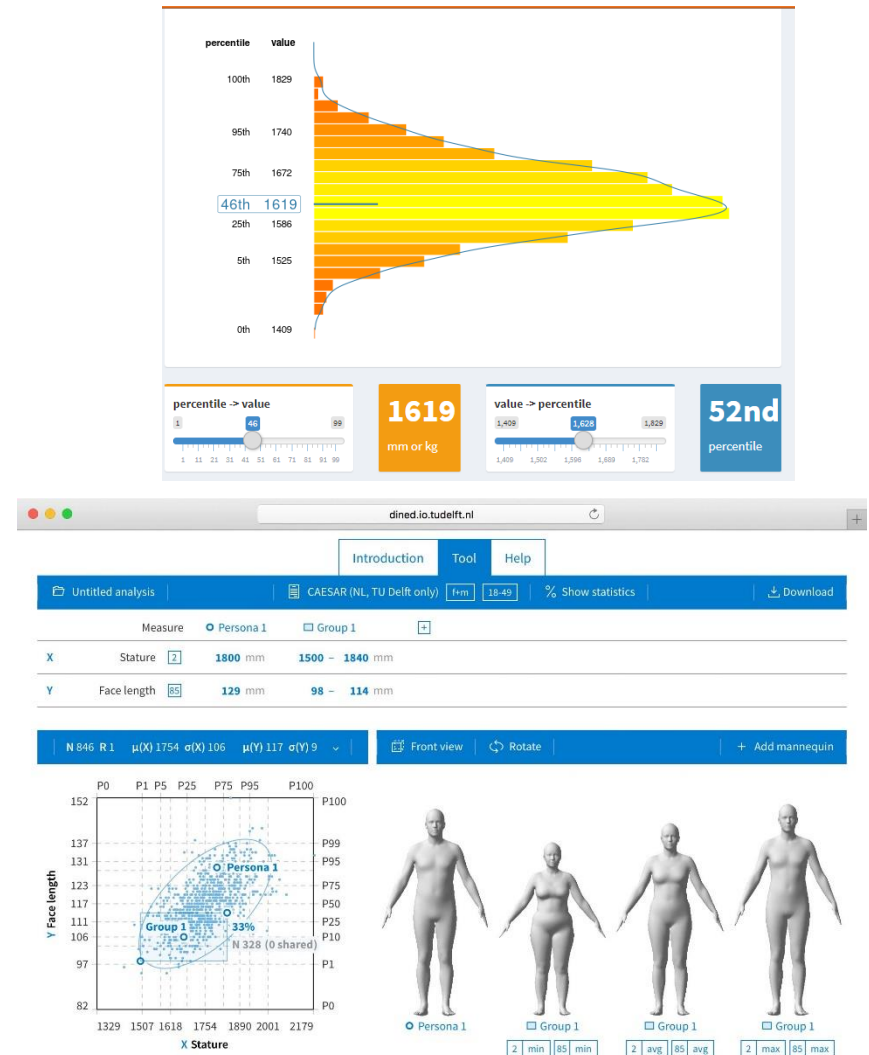
Open Anthropometric Databases

- ANSUR II

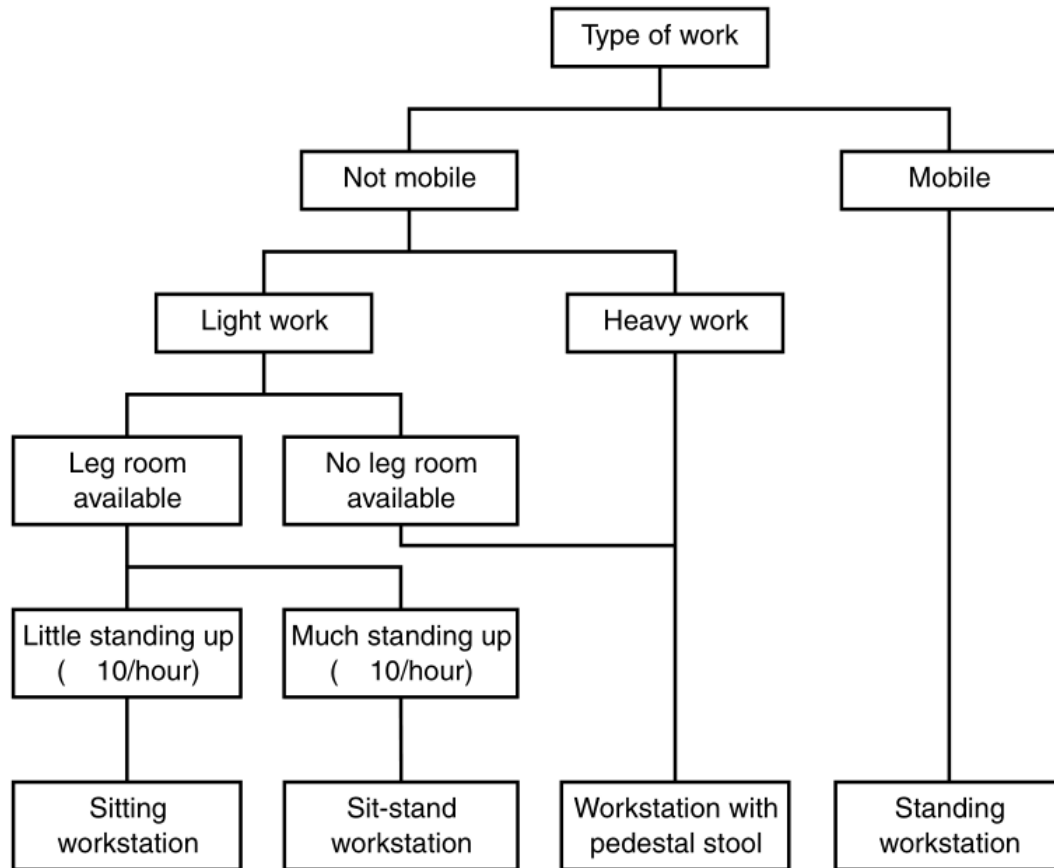
- The Anthropometric Survey of US Army Personnel (ANSUR 2 or ANSUR II), published internally in 2012 and made available publicly in 2017.

- DINED

- Project of the Applied Ergonomics and Design Section of the Human-Centered Design Department of the TU Delft faculty of Industrial Design Engineering.



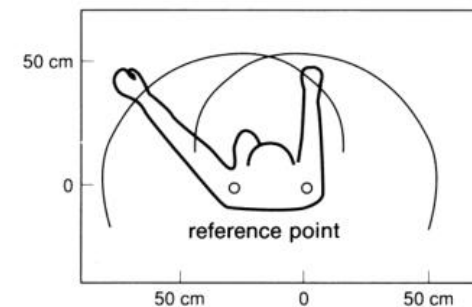
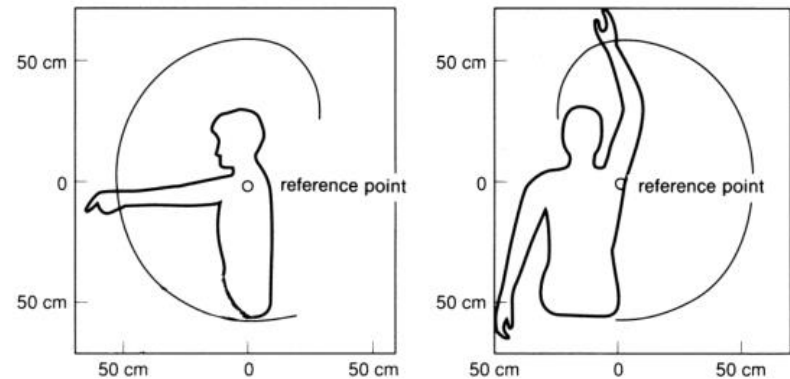
Selecting Posture



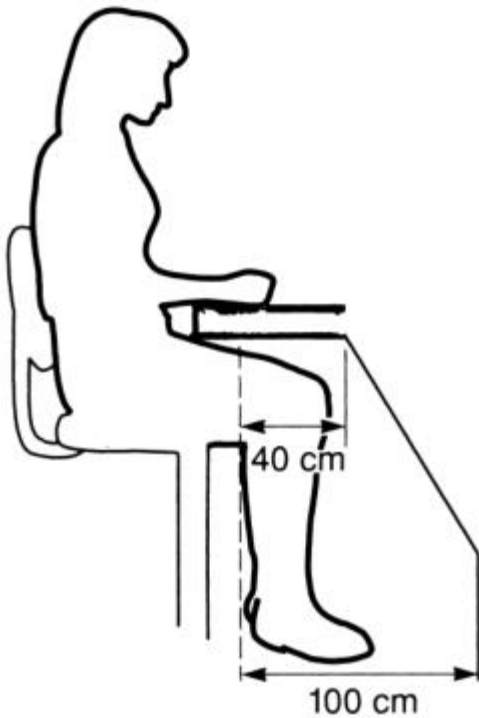
Postures: Sitting

- Sitting has advantages over standing because of several support surfaces:
 - Floor, seat, back rest, armrest, work surface
 - Good for activities which require less force
- Ergonomic Chairs:
 - Adjustable height: 39 – 52 cms
 - Adjustable backrest height: 20-30 cms

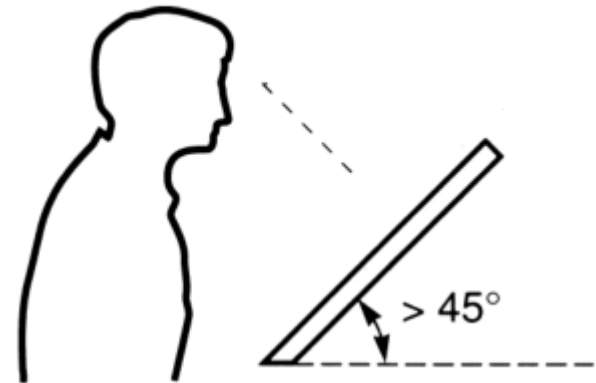
Type of task	Work height
Use of eyes: frequent, Use of hands/arms: infrequent	10–30 cm below eye height
Use of eyes: frequent, Use of hands/arms: frequent	0–15 cm above elbow height
Use of eyes: infrequent, Use of hands/arms: frequent	0–30 cm below elbow height



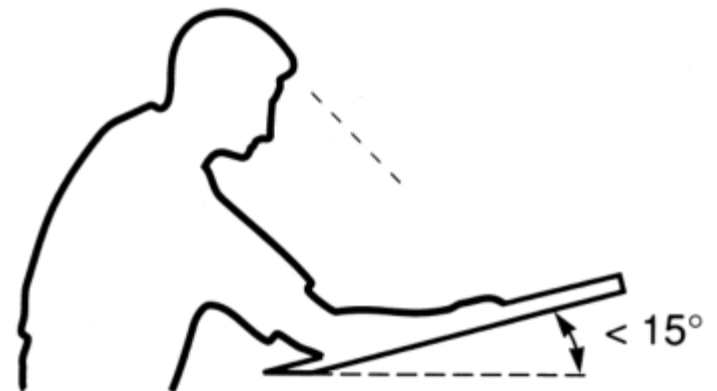
Postures: Sitting



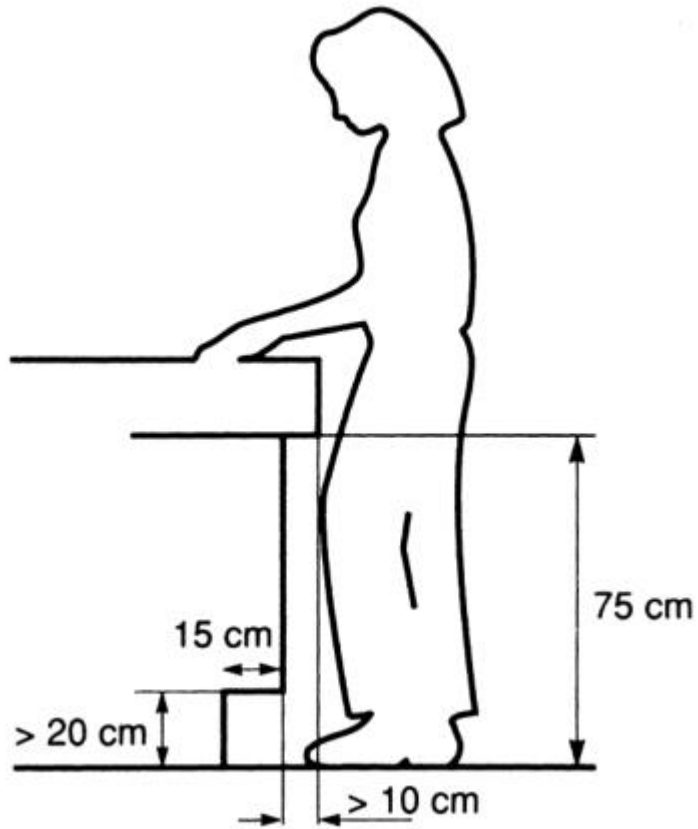
No manual work



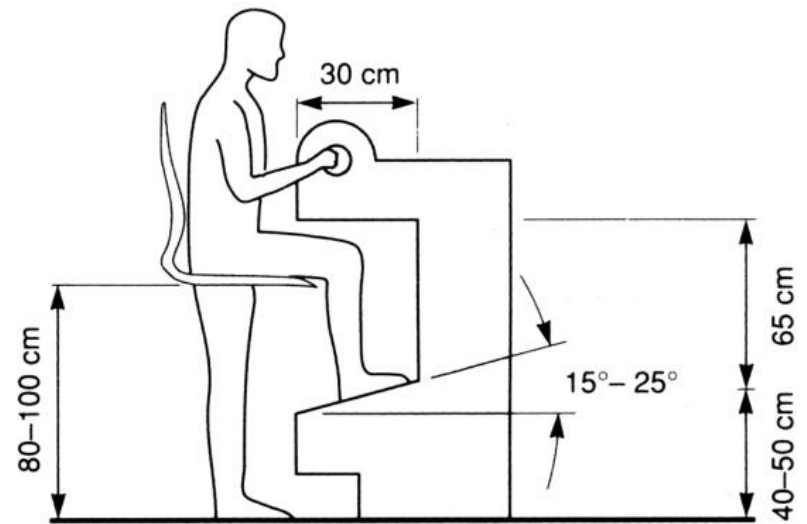
Manual work



Postures: Standing

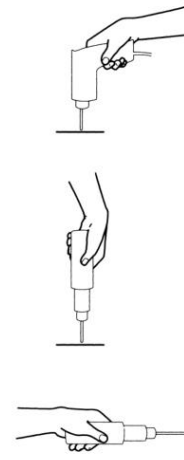


Alternate sitting and standing postures

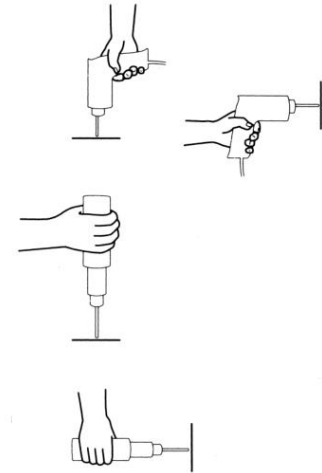


Hand-held tools

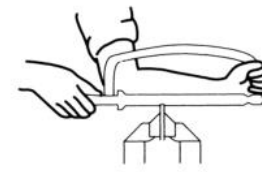
- When using hand-held tools, the wrist should be kept as straight as possible
- If activity requires, bending wrists, use curved tools
- Hand-held tools should weigh less than 2 kgs if cannot rest on a surface



(a) incorrect



(b) correct



(a) incorrect

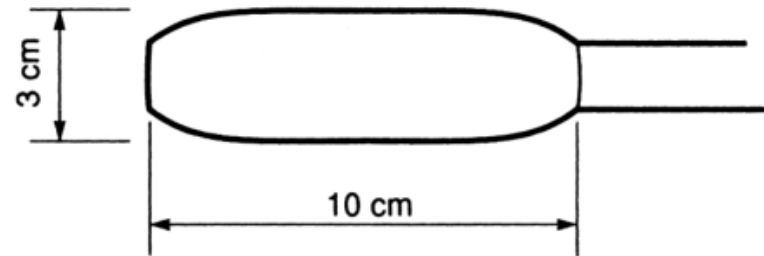


(b) correct



Hand-held tools

- Handgrip must have a diameter of 3 cms, and length 10 cms.
- Pre-shaped grips do not account for differences in finger size or gloves.
- Two types of grip:
 - **Power grip** – uses strong muscles in the forearm
 - **Precision grip** – uses smaller and weaker finger muscles, held between index and thumb fingers

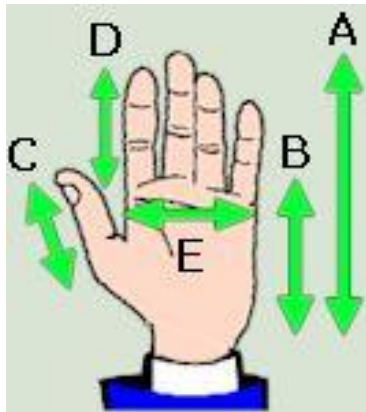


Badly Designed Hand-held Tools

- Repeated muscle use and excessing bending makes hands and wrists uncomfortable
- Wrist can be bent in 3 directions:
 - Backwards – extension
 - Forwards – flexion
 - Sideways – deviation (**dangerous!**)



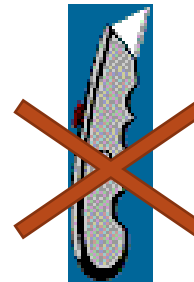
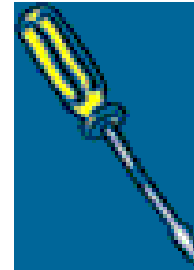
Anthropometry for hand tools



Anthropometric estimates for British adults aged 19-65 years (in mm)			
Dimension	5th %ile	50th %ile	95th %ile
A Hand length	173	189	205
	159	174	189
B Palm length	98	107	116
	89	97	105
C Thumb length	44	51	58
	40	47	53
D Index finger length	64	72	79
	60	67	74
E Hand breadth	78	87	95
	69	76	83
F Maximum grip diameter	45	52	59
	43	48	53

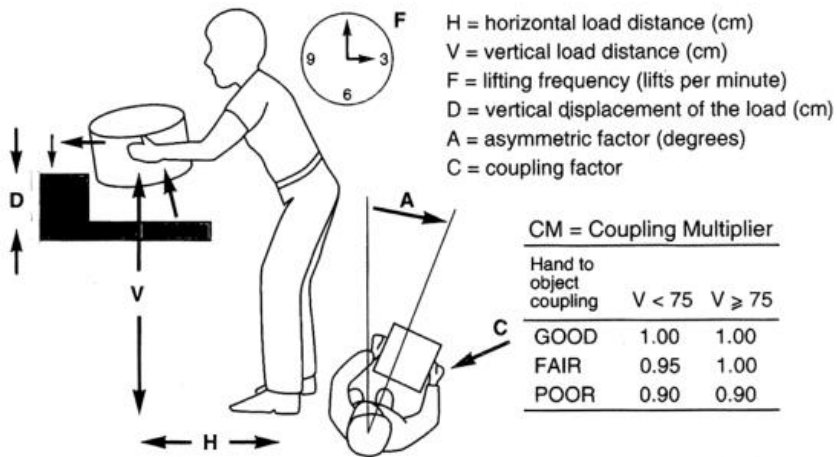
Guidelines Hand-held Tools

- Handle diameter
 - 30-45mm for power grip
 - 8-16mm for precision grip
- Handle length
 - 100-130mm, depending on tool size
- Handle shape
 - hexagonal/fluted with ridges
 - guard to block sharp parts



Movements:

Lifting and Carrying (less than 23 kgs)



NIOSH Lifting Calculator

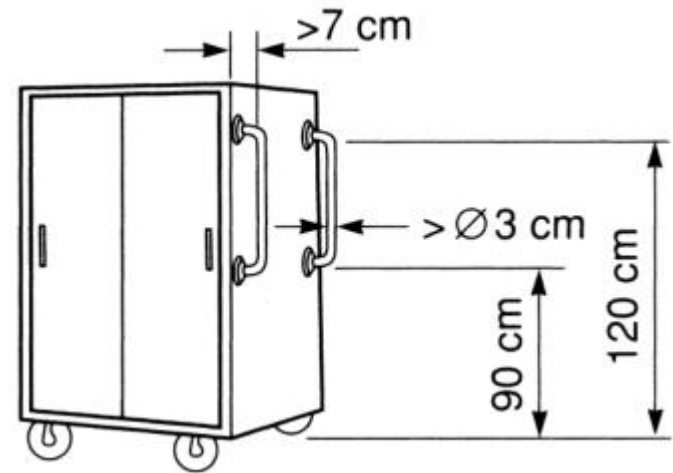
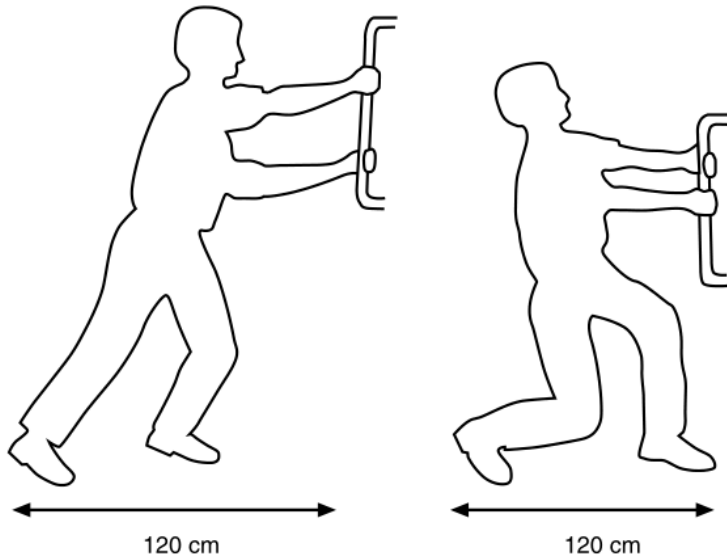
	Origin	Destination
Horizontal Location ⓘ	50	60
Vertical Location ⓘ	50	60
Travel Distance ⓘ	10	30
Angle of Asymmetry ⓘ	0	0
Coupling ⓘ	Good	Good
Frequency ⓘ	3	3
Avg. Load ⓘ	5	5
Max Load ⓘ	13	13
Duration ⓘ	Short (1 hour)	Short (1 hour)

Calculate

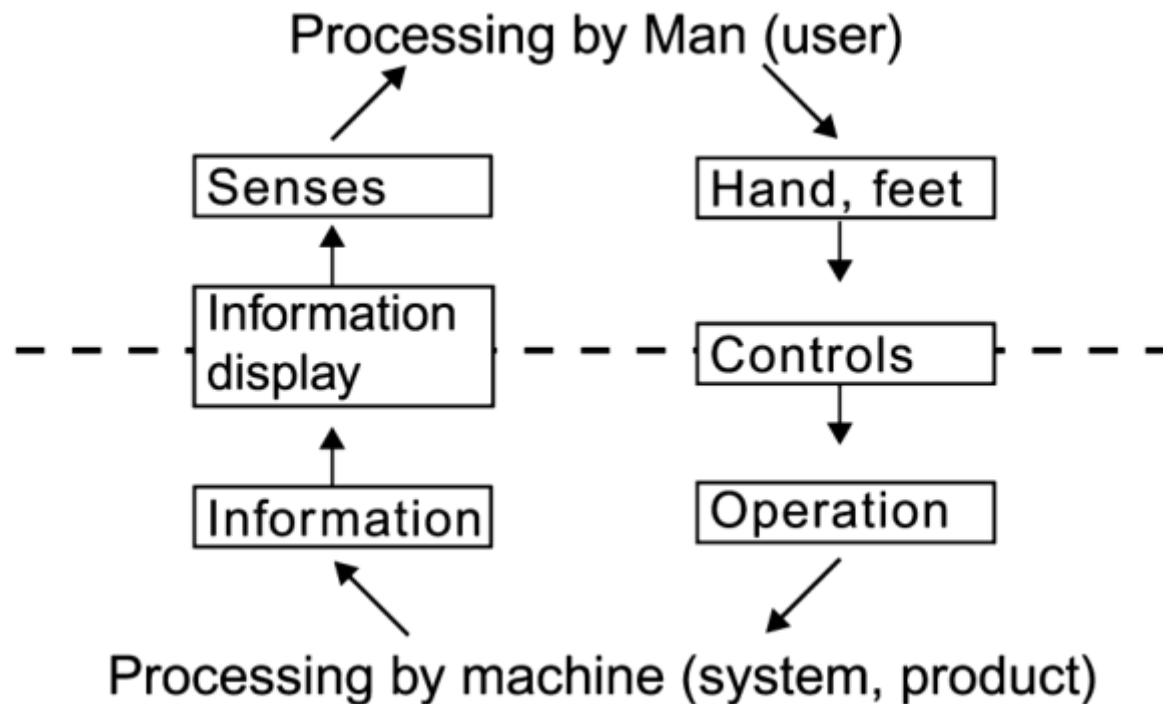
Clear **Print** **Save**

Movement:

Pulling/Pushing (less than 20 kg)

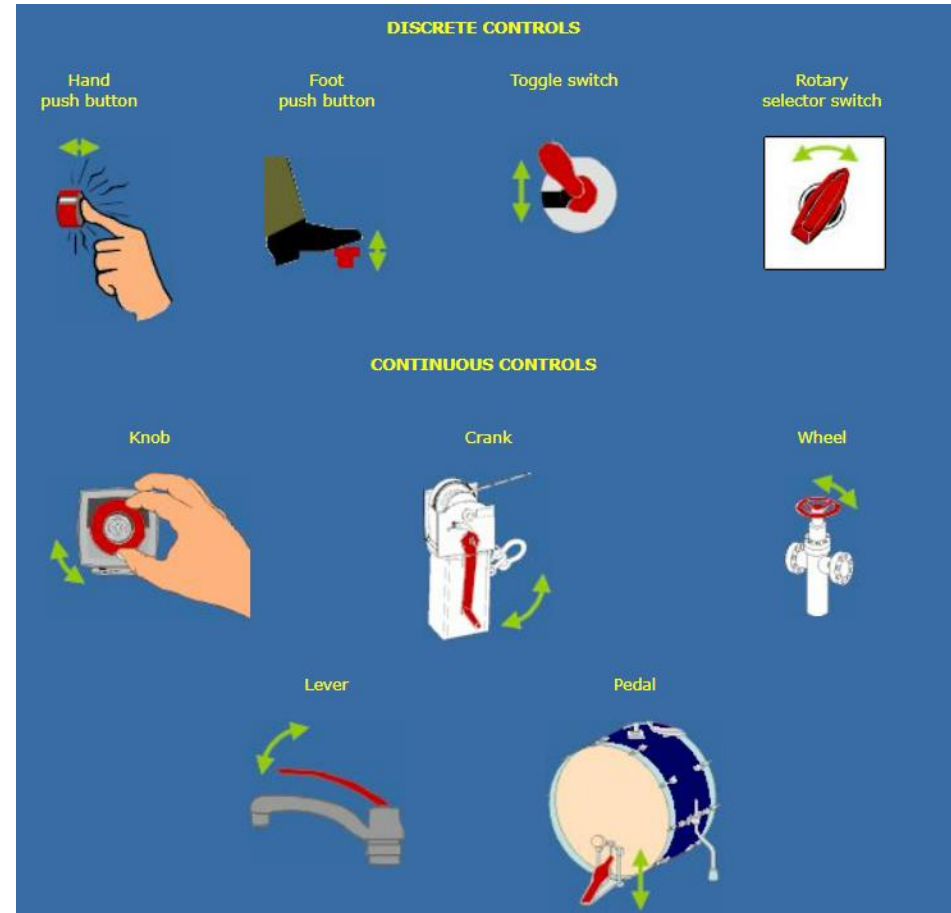


Operation and Information



Controls

- A control is a device that allows you to 'communicate' with objects and to manipulate them.
- The type of information that is transmitted by a control can be **discrete** or **continuous**.
- Some controls need less force, while others more depending on task

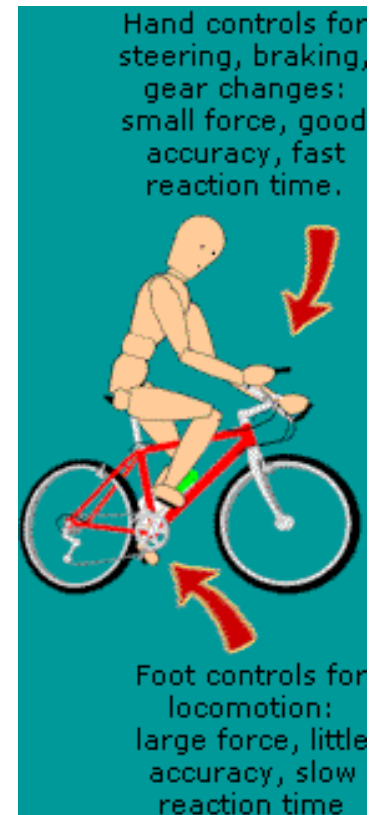
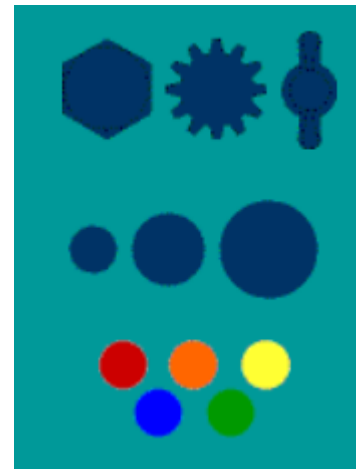
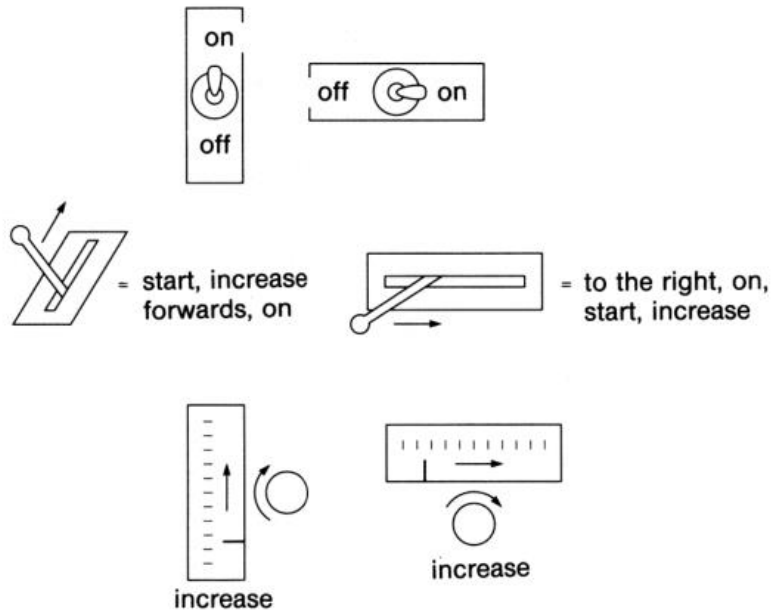


Effective Controls

- **Accessibility** - since almost all controls require physical contact to be operated, they must be able to be reached comfortably and efficiently.
- **Identifiability** - a control needs to be able to be identified in terms of what it does in order to operate (twist, lift etc.), what it controls and what state it is in (on, off, partially open, etc.).
- **Usability** - a control can be operated with the required force, speed and accuracy.

Guidelines for Control Design

- Control Compatibility
- Colour/shape/size coded

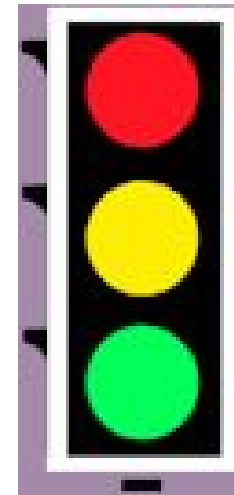


Choosing a control

Control type	Suitability for tasks involving:			
	Speed	Accuracy	Force	Range
Push button	Good	Unsuitable	Unsuitable	Unsuitable
Toggle switch	Good	Unsuitable	Unsuitable	Unsuitable
Rotary selector	Good	Good	Unsuitable	Unsuitable
Knob	Unsuitable	Fair	Unsuitable	Fair
Small crank	Good	Poor	Unsuitable	Good
Large crank	Poor	Unsuitable	Good	Good
Wheel	Poor	Good	Fair/Poor	Fair
Horizontal lever	Good	Poor	Poor	Poor
Vertical lever (to/from body)	Good	Fair	Short: Poor Long: Good	Poor
Vertical lever (across body)	Fair	Fair	Fair	Unsuitable
Joystick (lever)	Good	Good	Poor	Unsuitable
Pedal	Good	Poor	Good	Unsuitable

Displays

- Dials – scale and pointer, good for controlling processes and presenting analogue information
- Counters – show discrete information, good for reading
- Warning devices – show color-coded urgent information



What is Universal Design?

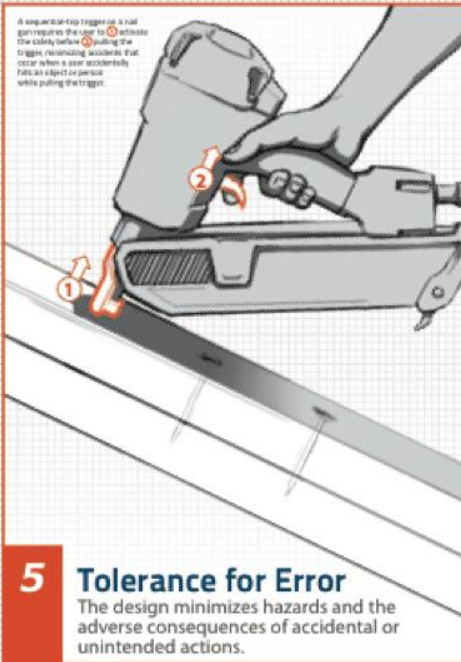
- Universal Design is an approach to design that honours human diversity.
- It addresses the right for everyone – from childhood into their oldest years – to use all spaces, products and information, in an independent, inclusive and equal way.
- It is a process that invites designers to go beyond compliance with access codes, to create excellent, people centred design.

The Principles of Universal Design



1 Equitable Use

The design is useful and marketable to people with diverse abilities.



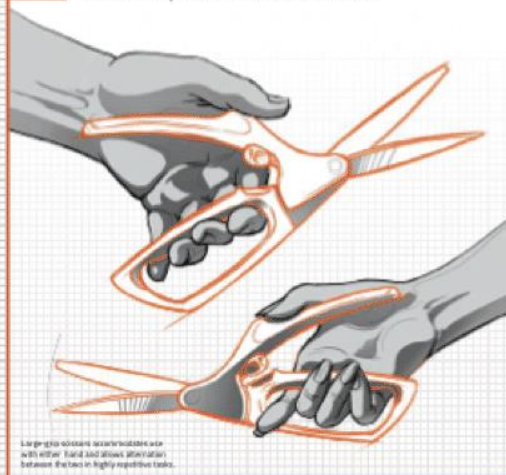
5 Tolerance for Error

The design minimizes hazards and the adverse consequences of accidental or unintended actions.

The design of products and environments to be usable by all people, to the greatest extent possible, without the need for adaptation or specialized design.

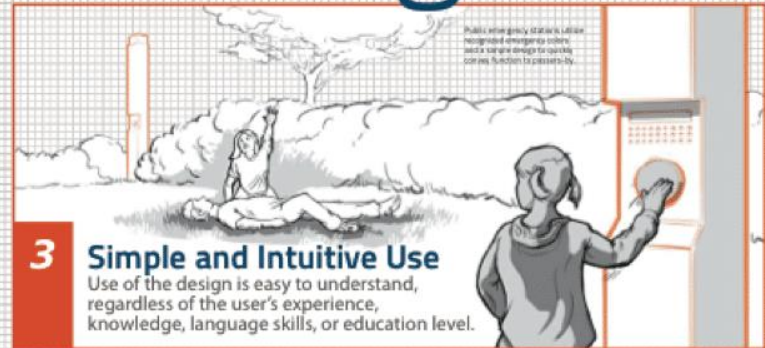
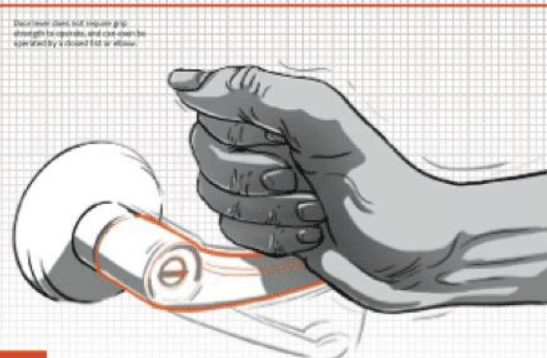
2 Flexibility in Use

The design accommodates a wide range of individual preferences and abilities.



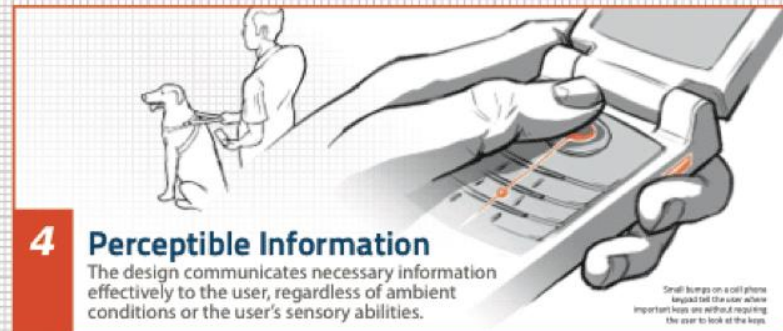
6 Low Physical Effort

The design can be used efficiently and comfortably and with a minimum of fatigue.



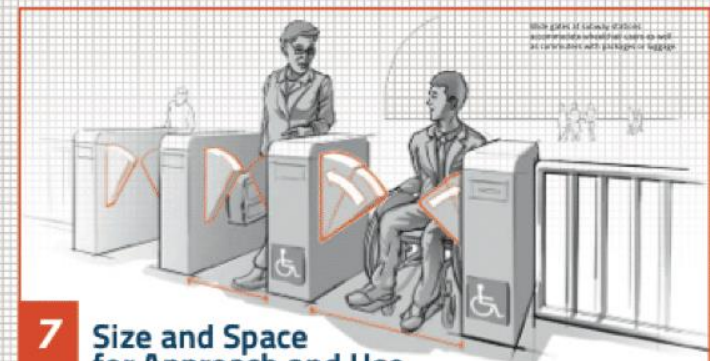
3 Simple and Intuitive Use

Use of the design is easy to understand, regardless of the user's experience, knowledge, language skills, or education level.



4 Perceptible Information

The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.



7 Size and Space for Approach and Use

Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.

The Principles of Universal Design

1. Equitable Use
2. Flexibility in Use
3. Simple and Intuitive
4. Perceptible Information
5. Tolerance for Error
6. Low Physical Effort
7. Size and Space for Approach and Use

Principle 1: Equitable Use

- The design is useful and marketable to any group of users
 1. Provide the same means of use for all users: identical whenever possible; equivalent when not
 2. Avoid segregating or stigmatising any users
 3. Provisions for privacy, security and safety should be equally available to all users
 4. Make the design appealing to all users
- Example:
 - A desk surface may be raised or lowered to accommodate users of varying height, or an individual who uses a wheelchair.

Can everyone use the same entrance?



Does the design provide the same means of use for all?



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Principle 2: Flexibility in Use

- The design accommodates a range of individual preferences and abilities
 1. Provide choice in method of use
 2. Accommodate right-handed or left-handed access and use
 3. Facilitate the user's accuracy and precision
 4. Provide adaptability to the user's pace
- Example:
 - A captioned video will allow people to choose to listen or to read in order to understand content.
 - This not only provides access to individuals with hearing impairments but also accommodates those who would rather not use sound or who comprehend better through reading.

Does the design provide choice in method of use?



Does the furniture adapt to people's needs and abilities?



Can the design be used by left and right handed people?



Principle 3: Simple and Intuitive

- Use of the design is easy to understand regardless of the user's experience, knowledge, language skills, or current concentration level.
 1. Eliminate unnecessary complexity
 2. Be consistent with user expectations and intuition
 3. Accommodate a wide range of literacy and language skills
 4. Arrange information consistent with its importance
 5. Provide effective prompting and feedback during and after task completion
- Example:
 - a website that is well-organized with clear headings will facilitate access to information.

Is it easy to understand? Can you make it work?



Principle 4: Perceptible Information

- The design communicates necessary information effectively to the user, regardless of ambient conditions or the user's sensory abilities.
 1. Use different modes (pictorial, verbal, tactile) for redundant presentation of essential information
 2. Provide adequate contrast between essential information and its surroundings
 3. Maximize 'legibility' of essential information and its surroundings
 4. Differentiate elements in ways that can be described (i.e. make it easy to give instructions or directions)
 5. Provide compatibility with a variety of techniques or devices used by people with sensory limitations
- Example:
 - a video includes a voiceover for individuals with visual impairments

Does the design use different modes for presentation?



Does the environment help you find your way?



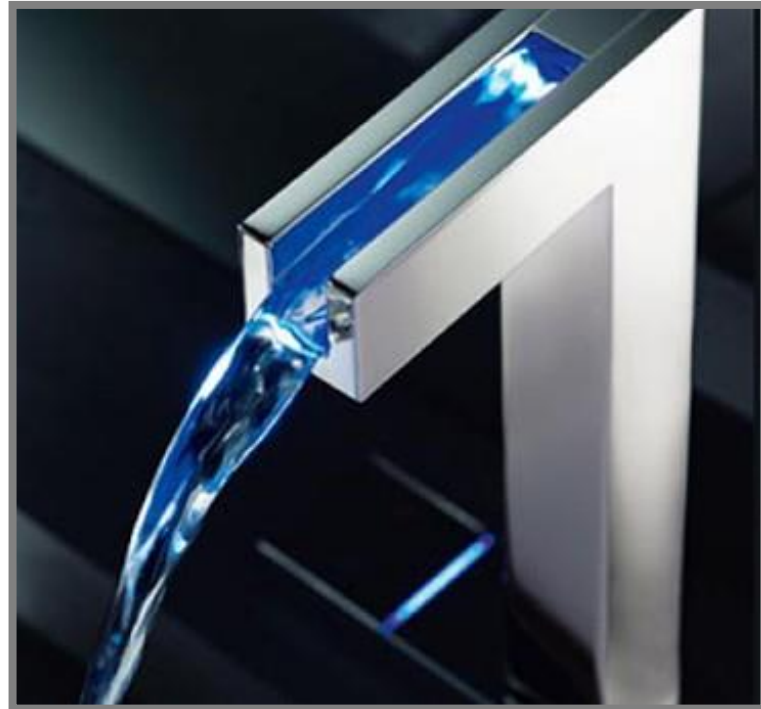
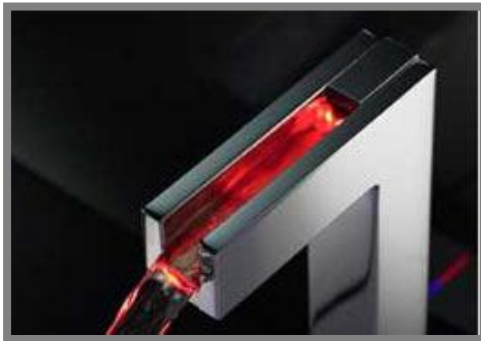
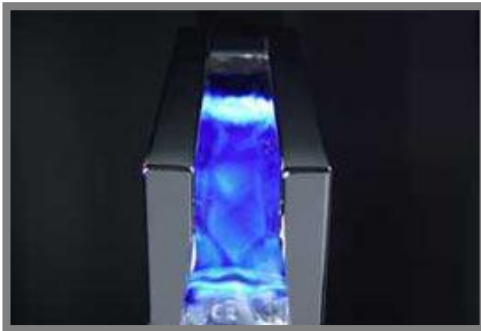
Principle 5: Tolerance for Error

- The design minimises hazards and the adverse consequences of accidental or unintended actions.
 1. Arrange elements to minimise hazards and errors: most used elements, most accessible; hazardous elements eliminated, isolated or shielded
 2. Provide warnings of hazards and errors
 3. Provide fail safe features
 4. Discourage unconscious action in tasks that require vigilance
- Example:
 - a hallway is free of protruding objects at a height where they would not be detectable by someone with a visual impairment who uses a cane.

Are there unexpected level changes?



How can you tell when the water is hot?



Principle 6: Low Physical Effort

- The design can be used efficiently and comfortably and with a minimum of fatigue
 1. Allow user to maintain a neutral body position
 2. Use reasonable operating forces
 3. Minimise repetitive actions
 4. Minimise sustained physical effort
- Example:
 - an automatic door opener can facilitate access to an office space or classroom.

Does the design help minimise the effort needed?



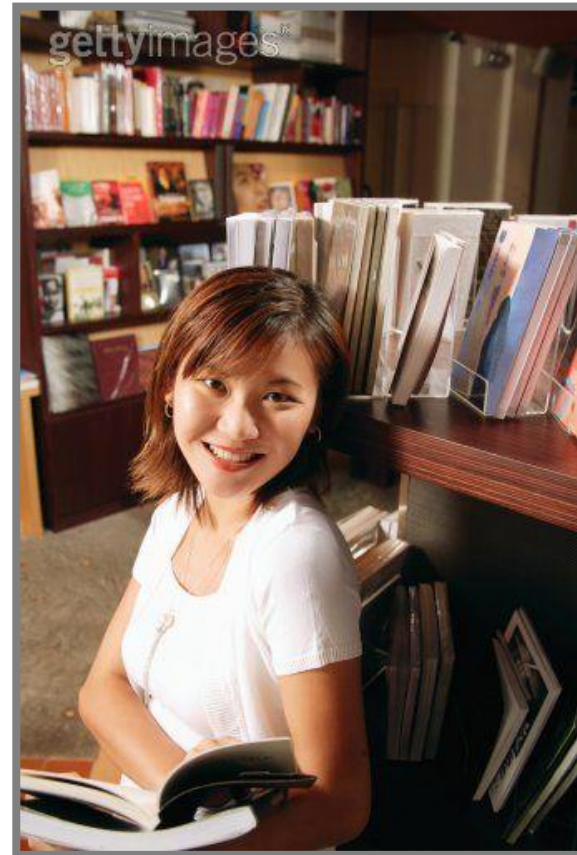
Principle 7: Size and Space for Approach and Use

- Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.
 1. Provide a clear line of sight to important elements for any seated or standing user
 2. Make reach to all components comfortable for any seated or standing user
 3. Accommodate variations in hand and grip size
 4. Provide adequate space for the use of assistive devices or personal assistance
- Example:
 - a classroom includes a range of seating options, including a table for someone who uses a wheelchair or wider chairs for individuals who are taller and/or larger.

Is there room to manoeuvre?



Is there space to examine the art exhibit?



Getting Stated with Fusion 360

Useful Resources

- Ergonomics Book
 - Dul, J., & Weerdmeester, B. (2008). *Ergonomics for beginners: a quick reference guide*. CRC press.
- Fusion 360 video tutorials
 - [Product Design Course using Fusion 360](#)
 - [Official Autodesk Tutorials](#)
- 3D models library:
 - www.thingiverse.com
 - www.youmagine.com
 - www.myminifactory.com